

Response to the Referees — Second Revision

Manuscript LR20628MR — “Yukawa screening derivation of the bond-valence rule”

We thank both referees. The First Referee, whose first-round report shaped the previous revision, now recommends publication. The Second Referee is new to the manuscript and raises concerns in four areas. These are the alternative empirical forms of the bond-strength law and the notation of the expansion point, the construction and interpretation of manuscript Fig. 1 and the Supplemental figures of B – R_0 lines, the physical motivation for expanding about the Gibbs mean in the partition-thermodynamics section, and the transferability of the bond-valence parameters when each structure is fitted individually. We respond to each point below.

Summary of changes. Five passages of the manuscript were changed in response to the Second Referee, all in the main text and all marked in the accompanying tracked changes. First, the sentence after Eq. (3) was reworded to remove the ambiguity the referee read as a typographical error (Block 3a). Second, a clause after Eq. (4) now identifies B as the inverse log-weight slope at the expansion point, and the sentence introducing Eq. (5) states that the uniform-limit relation follows from Eq. (1) alone (Block 3b). Third, a statement after Eq. (5) makes explicit that the predicted slope describes the family of per-structure fits rather than prescribing a single pair (Block 6). Fourth, the Partition thermodynamics section closes with the physical argument for expanding about the Gibbs mean (Block 8). Fifth, the paragraph after Eq. (4) now places the Brown–Shannon power law as the same screened weight linearized in $\ln R$, with four references added (Block 2). Two figures were prepared for this response. Response Figure 1 shows the distribution of the per-structure screening length across the corpus, and Response Figure 2 redraws the Co–O panels of Supplemental Fig. S4 with the underlying per-structure points. Every statistic quoted in this letter is reproduced by scripts in the project repository [1]. In the Supplemental Material, the fifteen coordination-resolved B – R_0 line figures (Figs. S2–S16) were regenerated to show the per-structure points within a physical window, with the convention stated in the text, and the Data provenance section now states the stability filter.

Second Report of the First Referee

“The authors have substantially improved their manuscript and I can now recommend publication.”

Response. We thank the referee for the careful first-round report and for this recommendation.

Report of the Second Referee, with responses

Block 1 — introductory remarks.

“The manuscript proposes a theoretical basis for calculating bond strength in bond-valence theory, deriving the bond strength from the electrostatic flux associated with a Yukawa potential. Bond-valence theory is an empirical framework widely used in materials science, particularly for the validation of crystal structures and the study of ion

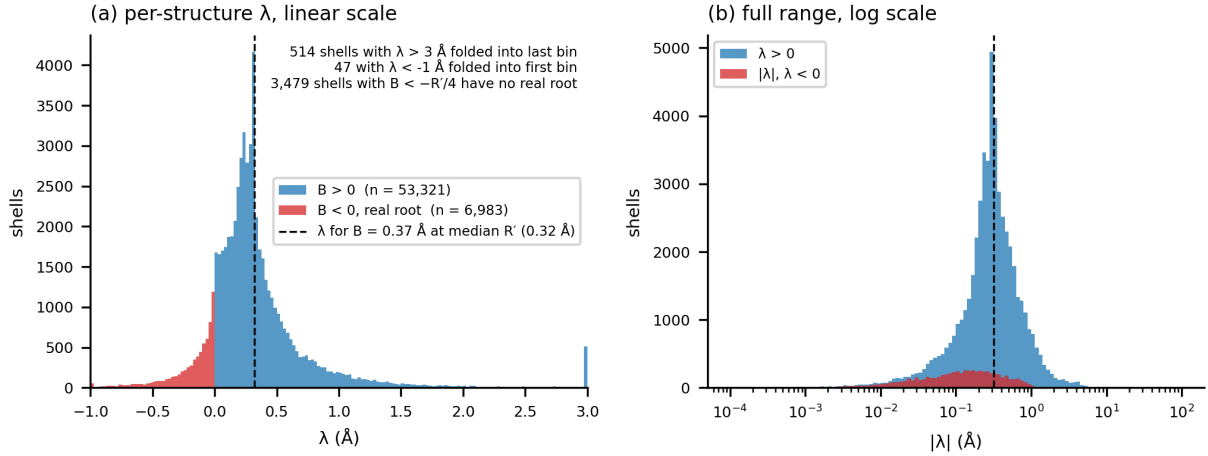
diffusion in solid-state ionic conductors. I believe that this could be an interesting topic for the readers of Physical Review Materials.

However, I did not find the manuscript particularly convincing. The exposition is not always clear, some aspects of the theoretical framework are not sufficiently justified, and I have serious concerns about the validity of some of the calculations presented, as well as about the interpretation of some of the figures.”

Response. We thank the referee for the assessment of the topic and for a report whose specific concerns we can act on.

The referee’s central worry, developed in Blocks 5 through 7 and in the closing remarks, is that the per-structure (R_0, B) values span ranges that appear unconventional. We clarify the relationship between the bond-valence convention and its physical underpinning. Although (R_0, B) are conventionally read as a reference bond length and a softness length, no prior theory bounds the values they may take.

Our position is that bond valence theory is an approximation, and the application of it to a large corpus of structures will identify materials that do not conform to the assumptions of the approximation. The conventional approach has often relied on carefully curated data sets from which non-conforming structures were removed. For example, one seminal study eliminated about 15% of the alkali coordinations in its reference set, relative to the 1,524 retained for the final refinements, because of their “strange bond-valence sums” [2]. This carries a risk of circularity, since selecting structures that conform to the theory is not a test of its applicability, and some of the structures whose per-structure fits look “strange” are very common ground-state ones, such as the nearly degenerate SiO_2 polymorphs discussed in the manuscript’s screening-collapse section. Here,



Response Figure 1: Per-structure Yukawa screening length $\lambda = (-R' + \sqrt{R'^2 + 4BR'})/2$, with R' the shell’s mean bond length, for all 63,783 single-valence fitted cation–oxygen shells in the corpus. (a) Linear scale from -1 to 3 Å. Blue, the screening branch $B > 0$ (53,321 shells). Red, the anti-screening branch $B < 0$ where the root is real (6,983 shells). The 514 shells with $\lambda > 3$ Å and 47 with $\lambda < -1$ Å are folded into the end bins, and the 3,479 shells with $B < -R'/4$ have no real root. The dashed line is λ for the conventional $B = 0.37$ Å at the corpus median bond length of 2.02 Å. (b) The same values as $|\lambda|$ on a logarithmic axis over the full range.

we instead force the theory to confront a large set of DFT-relaxed structures from the Materials Project, a database in wide community use, screened for thermodynamic stability as described under Block 7. Letting the poor fits mark the limits of the approximation is part of the result, and it points towards a physical basis for the parameters. Specifically, we test bond valence theory against DFT and show exactly where it fails, pointing to specific failure mechanisms (lone pairs, Jahn–Teller physics) rather than omitting failures from the fit and leaving their status ambiguous.

The new physical content we present is the screening length, and it does not suffer from the extreme values that the per-structure (R_0, B) can take. Response Figure 1 shows the per-structure $\lambda = (-R' + \sqrt{R'^2 + 4BR'})/2$, with R' approximated here as the mean bond length of the shell, for all 63,783 single-valence fitted shells in the corpus. On the screening branch ($B > 0$, 84% of shells) λ is a single peak with median 0.30 Å and interquartile range 0.18–0.47 Å, 95% of the values lie below 1.2 Å, and only 1% exceed 3 Å. Along the degenerate direction of the (R_0, B) plane the inversion of Eq. (4) gives $\lambda \simeq \sqrt{BR'}$, so a B of 10^3 Å becomes a λ of about 45 Å and the tail is compressed by two orders of magnitude. The conventional $B = 0.37$ Å corresponds to $\lambda = 0.32$ Å at the median bond length, at the center of the peak. The anti-screening branch ($B < 0$) is 16% of shells. Of these, 11% of all shells have a real root, with median $\lambda = -0.12$ Å, and 5% have $B < -R'/4$ and no real root, which marks where the first-order Yukawa form ceases to apply. The species-level λ^* tabulated in the Supplemental Material (median 0.31 Å, interquartile range 0.26–0.37 Å) sit at the center of this per-structure distribution.

Block 2 — alternative empirical forms of the bond-strength law.

“The derivation of the bond strength from a Yukawa potential is interesting. It is an extension of a previous idea that linked electrostatic flux with bond strength. This work shows that when the electrostatic flux is derived from a Yukawa potential, the empirical exponential form of the bond strength is recovered. However, there are other empirical forms that also work well, such as the original power-law relationship introduced in the original formulation of bond-valence theory by Brown and Shannon (Acta Cryst. A29, 266–282, 1973). The manuscript neither mentions nor discusses these alternative formulations, which give different dependencies of the bond strength on the bond distance.”

Response. The referee is right that the power-law form has been used previously in the literature. Brown and Shannon [3] fitted individual curves $S = s_0(R/R_0)^{-N}$ and universal curves $S = (R/R_1)^{-N}$, with R_1 the length of a bond of unit valence and N between 4.1 and 6.1 for the isoelectronic cation series, building on the per-polyhedron power-law curves of Donnay and Allmann [4]. Brown and Wu [5] tabulated (R_1, N) for cation–oxygen pairs, and the power law was in routine use through the 1970s. Zachariasen [6] compared it with the logarithmic form $D(s) = D(1) - B \ln s$, which is Eq. (1) solved for the bond length, and found that the two agree to better than 1% for $0.5 < s < 2$ when $N = D(1)/B$, but differ by 5–7% in bond length at $s = 0.1$.

In fact, the two forms are not in competition within the limits of the approximation. Both are leading-order expansions of the same Yukawa log-weight, taken in different variables. Expanding $\ln w_{ij}$ in R_{ij} about R' gives the exponential of Eq. (1) with $B = \lambda(\lambda + R')/R'$, which is Eq. (4). Expanding the same $\ln w_{ij}$ in $\ln R_{ij}$ about $\ln R'$ gives the power law, because

$$\frac{d \ln w}{d \ln R} = -\frac{(R/\lambda)^2}{1 + R/\lambda}, \quad \text{so} \quad N = \frac{R'^2}{\lambda(\lambda + R')} = \frac{R'}{B}.$$

This is Zachariassen’s condition $N = R_1/B$ for the two forms to coincide [6], obtained here as the ratio of the two leading-order expansions of one screened weight. With $B \simeq 0.37 \text{ \AA}$ and the Brown–Shannon $R_1 \simeq 1.4\text{--}1.8 \text{ \AA}$ it gives $N \simeq 4\text{--}5$, in the range of their universal exponents. The exponential is the form that is linear in the bond length, which is why it is the one whose $B\text{--}R_0$ correlation has a parameter-free slope, but the power law is the same screened weight linearized in the logarithm of the bond length.

We retain the exponential form because it is linear in the bond length, which is what gives the $B\text{--}R_0$ correlation its parameter-free slope.

Changes. The manuscript now discusses the power law. After Eq. (4) it states: “The power-law form $S = (R/R_1)^{-N}$ of Brown and Shannon [3], which traces to Donnay and Allmann [4] and was tabulated for cation–oxygen pairs by Brown and Wu [5], is the same screened weight linearized in $\ln R$ instead of R , with exponent $N = R'/B$. This is the condition under which Zachariassen found the two forms to agree to within 1% for $0.5 < S < 2$ [6]. We retain the exponential because it is linear in the bond length, which is what gives the $B\text{--}R_0$ correlation below a parameter-free slope.” The four references were added to the manuscript bibliography.

Block 3 — minor remark on R' versus R_0 . The referee’s remark contains two questions, answered separately.

Block 3(a) — is “at $R_{ij} = R'$ ” a typographical error?

“A minor remark. After linearizing the weights (Eq. 3), the authors state: ‘Exponentiating the linear term and renormalizing therefore reproduces Eq. 1 at $R_{ij} = R'$.’ I suspect this is a typographical error, and that the authors actually mean $R' = R_0$ because this is the choice that reproduces Eq. 1.”

Response. It is not a typographical error, but the sentence was ambiguous and we have reworded it to improve clarity. Exponentiating the linear term of Eq. (3) and renormalizing reproduces the exponential form of Eq. (1) for every bond in the shell, whatever the expansion point. What the expansion point sets is where the linearized weight equals the exact Yukawa weight. The phrase “at $R_{ij} = R'$ ” meant that the error δ_{ij} vanishes at that bond length and grows quadratically away from it. It was not a statement about R_0 . The choice $R' = R_0$ would not make Eq. (1) exact anywhere in particular, because R_0 is in general not a bond length of the shell. It is the intercept of Eq. (1), and the partition identity of Eq. (10) shows that it differs from the valence-weighted shell centroid m_i by the entropy correction $B(\ln q_i - H_i)$, which in the uniform-bond limit is $B \ln(z/n)$. The two lengths coincide only for a single bond of unit valence, the Brown limit noted after Eq. (10).

Changes. The sentence after Eq. (3) now reads “Exponentiating the linear term and renormalizing reproduces the exponential form of Eq. (1). The log-weight error vanishes at $R_{ij} = R'$ and is quadratic away from it.”

Block 3(b) — why R' appears in Eq. 4, and whether Eq. 5 assumes $R' = R_0$.

“Why is R' used in Eq. 4 rather than R_0 ? In Eq. 5, the authors calculate dB/dR_0 from Eq. 4, which appears to implicitly assume that the R' in Eq. 4 is in fact R_0 .”

Response. R' appears in Eq. (4) because B is defined there as the inverse of the local slope of the Yukawa log-weight, $-d \ln w / dR = R / [\lambda(\lambda + R)]$, evaluated at the expansion point. That

slope depends on where in the shell the expansion is centered, so the composite $B = \lambda(\lambda + R')/R'$ necessarily carries R' . Replacing R' by R_0 would evaluate the slope at a length that is not in the shell.

Equation (5) is not derived from Eq. (4) and makes no use of R' , so no identification $R' = R_0$ is assumed. It follows from applying the valence-sum rule to Eq. (1) in the uniform-bond limit, which gives $R_0 = \bar{R} + B \ln(z/n)$, and differentiating at fixed $\bar{R}$. The uniform-bond limit also shows directly that R' and R_0 differ. With all $R_{ij} = \bar{R}$ the expansion point is the shell radius, $R' = \bar{R}$, while $R_0 = \bar{R} + B \ln(z/n)$ equals it only at $z = n$. Equation (4) enters only when the fitted B is read as a screening length, and there the manuscript states the operational choice of R' explicitly. Within a shell the partition-thermodynamics section fixes $R' = m_i$, the Gibbs mean.

Changes. The sentence introducing Eq. (5) now states that the uniform-limit relation follows from the valence-sum rule applied to Eq. (1), involves only the fitted parameters, and uses nothing from Eq. (4). A clause after Eq. (4) now identifies B as the inverse of the log-weight slope $-d \ln w / dR = R / [\lambda(\lambda + R)]$ evaluated at the expansion point R' .

Block 4 — construction of manuscript Figure 1.

“I do not clearly see how Figure 1 has been calculated. My understanding is the following. For a given A–O pair, the authors have analyzed several crystal structures containing A–O interactions, obtaining one pair (R_0, B) for each structure. When B is plotted against R_0 , a strong linear correlation is found for structures with the same coordination number. The slope of this linear correlation is $1/\ln(z/n)$. Figure 1 appears to have been constructed from the data shown in Figure S2. If I have understood this correctly, I have several concerns.”

Response. The referee’s understanding is correct, with two small refinements. Each DFT-relaxed structure contributes one jointly fitted (R_0, B) pair. For each species and coordination number n the pairs are regressed by an outlier-robust RANSAC estimator refit on its inliers, and the slope of that regression is one point of manuscript Fig. 1. The lines drawn in the Supplemental figures are these regressions, so manuscript Fig. 1 is built from the lines of Figs. S2–S16, not of Fig. S2 alone. The construction is described in the manuscript (Empirical validation) and in the Supplemental Material (“Line-fitting procedure”). The prediction of Eq. (5) is that the slope of the per-structure correlation is $1/\ln(z/n)$ with no fitted constants.

Block 5 — (a) range of the R_0 and B values in the Supplemental line figures.

“The ranges of R_0 and B shown in Figure 1 are extraordinarily large. One would normally expect R_0 to be of the same order of magnitude as the characteristic bond distance. Taking Co–O as an example, for $\text{Co}^{+1}\text{--O}$, R_0 appears to vary between approximately 0 and 200 Å; for $\text{Co}^{2+}\text{--O}$, between approximately –10 and 10 Å; and for $\text{Co}^{3+}\text{--O}$, between approximately –1000 and 1000 Å. These values do not appear physically meaningful. The parameter B presents a similar problem. In conventional bond-valence parameterizations, B is typically of the order of 10^{-1} Å. In the case of $\text{Co}^{3+}\text{--O}$, for example, the values shown in the figure appear to range from approximately –1000 to 1000 Å. Moreover, several of the curves extend into the region of negative B (anti-screening according to the formulation presented in the manuscript, so bonding forces are increasing with bond distance). ”

A further issue is that the individual (R_0, B) values obtained from the different crystal structures do not appear to be shown in Figure 1. Instead, only the least-squares regression lines are displayed. Why are the individual fitted (R_0, B) pairs not shown? I think it is important to show them explicitly, this would allow the reader to assess the actual distribution of the fitted parameters, the quality of the linear correlations, and the extent to which the regression lines are representative of the underlying data. I wonder whether these extraordinarily large ranges could result from an error in the way the curves are represented or plotted. If this is indeed the case, it would be rather surprising that such an issue was not identified during the preparation or checking of the figures, even though they are included in the Supplementary Material. The authors should therefore carefully verify the numerical values and the plotting procedure used to generate these curves.”

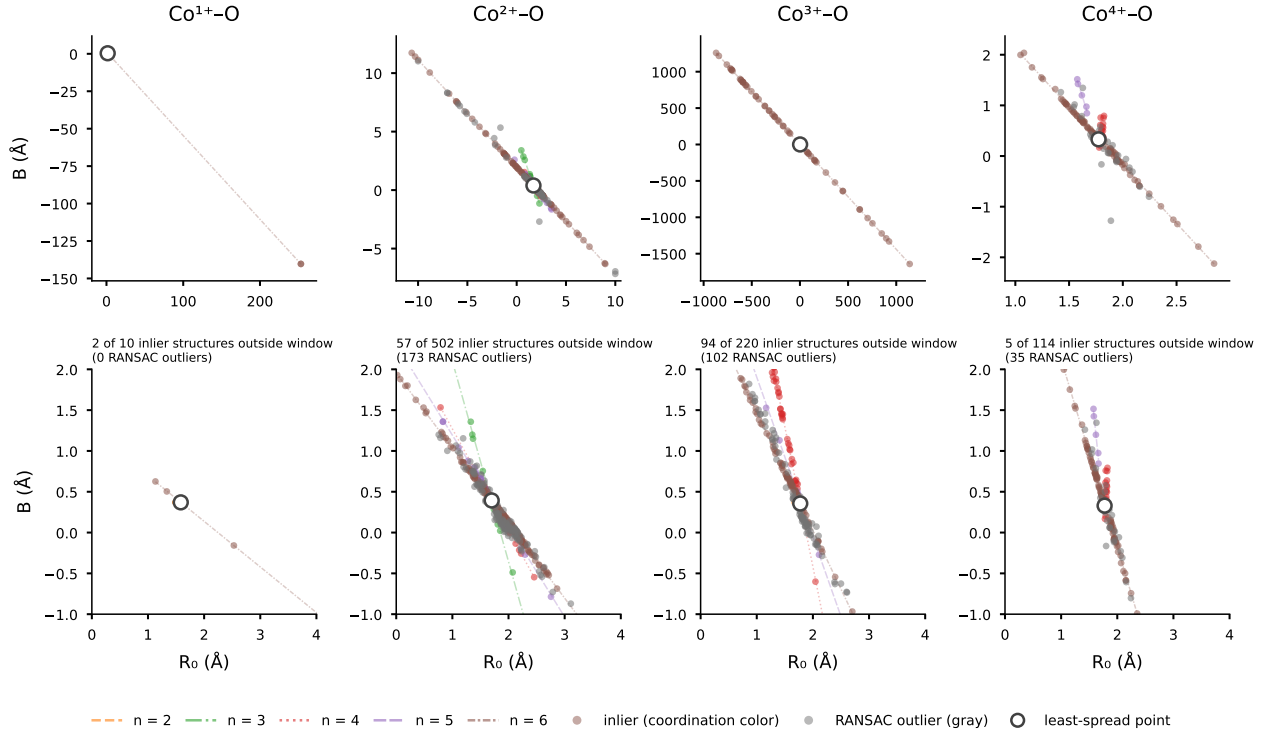
Response. We agree that there are extreme values that look surprising, and we regret that the original figures did not portray these values clearly, as they are portrayed in the preceding manuscript on the method [7]. Response Figure 2 reproduces the four Co–O panels of Supplemental Fig. S4 with the per-structure (R_0, B) pairs drawn under the fitted lines, both at the full range of the fitted values and within a restricted window, and all Supplemental line figures (Figs. S2–S16) have been regenerated with the per-structure points in the same window.

We note that the large majority of structures have (R_0, B) in conventional ranges. Of the 63,783 single-valence shells in the corpus, 92% lie within $0 < R_0 < 4$ Å and $-1 < B < 2$ Å, and 74% within $0 < R_0 < 4$ Å and $0 < B < 1$ Å. For the Co species of Response Figure 2 the in-window fractions are 80%, 89%, 67% and 93% for Co¹⁺ through Co⁴⁺. The 8% of shells outside the window is below the roughly 15% of coordinations that were eliminated from another reference set for “strange bond-valence sums” [2].

Values in the tails are not outliers in the sense of being poorly fitted or subject to numerical error. They are degenerate. For a nearly regular shell, Eq. (1) and the valence-sum rule constrain the joint fit of (R_0, B) only along the line $R_0 = \bar{R} + B \ln(z/n)$, and with equal bond lengths B is unconstrained altogether. The second-order terms δ_{ij} of Eq. (3) select a point on that line, but when the bond-length spread is a few hundredths of an ångström the selection is weak and the least-squares solution can lie along a wide range of the line, including at $|R_0|$ and $|B|$ of order 10^3 Å.

The data bear this out. The shells outside the window have a median bond-length range of 0.034 Å, against 0.107 Å for those inside, and for Co³⁺ at $n = 6$ the contrast is 0.023 versus 0.283 Å. Such a structure is an inlier precisely because it lies on the line, and a structure fitted at $(R_0, B) = (1000, 700)$ Å carries the same information about the slope as one fitted at $(1.9, 0.4)$ Å and almost none about a preferred point. The referee is right that the individual values are not meaningful as lengths. The negative- B segments arise in the same way, since a line drawn across the full inlier range crosses $B = 0$ wherever the inliers straddle the shell radius, which is where the line meets the axis by Eq. (6). The per-structure fits that return $B < 0$ are 16% of the corpus, dominated by near-degenerate shells, and are catalogued in the Supplemental sign census.

This fit degeneracy is equally present in conventional fitting, which suppresses it by fixing B to a constant [8] or by pooling many structures to select one point in the shallow valley of the residual surface [9]. We include every structure instead, not to assert that a B of 10^3 Å is a physically meaningful softness, but because such structures constrain the slope that Eq. (5) predicts as well as any other, and in order to be transparent about the implications of applying the exponential approximation universally.



Response Figure 2: The four Co–O panels of Supplemental Fig. S4 redrawn with the per-structure (R_0, B) pairs under the coordination-resolved lines. Top row, the full range of the fitted values, as plotted in the Supplemental Material. Bottom row, the same lines and points within $0 < R_0 < 4 \text{ \AA}$ and $-1 < B < 2 \text{ \AA}$, with the number of inlier structures outside the window stated above each panel. Every per-structure pair is a transparent filled circle, colored by coordination number for RANSAC inliers and gray for RANSAC outliers. Line styles encode the coordination number n , and the open circle is the least-spread intersection (R_0^*, B^*). Inlier and outlier counts are 10 and 0, 502 and 173, 220 and 102, and 114 and 35 for Co^{1+} through Co^{4+} . The far-out points on the $n = 6$ lines are shells with nearly equal bond lengths, for which the joint fit is degenerate along the line.

Changes. Response Figure 2 was prepared for this response by re-running the Supplemental line-fitting pipeline for the four Co species with the per-structure (R_0, B) pairs retained, so its lines are identical to those of Supplemental Fig. S4 (same seeded RANSAC fits, slopes and intercepts equal in all 13 species and coordination fits). All Supplemental line figures (Figs. S2–S16) have been regenerated to show the per-structure (R_0, B) pairs beneath the fitted lines, with RANSAC inliers as filled circles and outliers as open circles, axes fixed to $0 < R_0 < 4 \text{ \AA}$ and $-1 < B < 2 \text{ \AA}$, and the number of inlier structures outside the window stated above each panel. The fitted lines are unchanged (slopes and intercepts identical to the previous version for all 103 species). The Supplemental text introducing these figures now states the plotting convention and explains the out-of-window structures. The corpus-wide and Co-specific statistics quoted above are reproduced by two scripts added to the project repository [1].

Block 6 — (b) interpretation of the slope dB/dR_0 .

“From the uniform bond limit, $R = R_0 - B \ln(z/n)$, the authors derive the slope $dB/dR_0 = 1/\ln(z/n)$. They then use this equation to fit the data in Figure 1. If R_0 and B in the uniform bond limit equation are treated as variables, the above equation simply states that there are infinite pairs (R_0, B) that satisfy the valence-sum rule for a regular coordination polyhedron with a given bond distance and coordination number.

In my understanding, the uniform-bond limit is instead used to determine the constant parameters R_0 and B when a set of crystal structures with regular coordination polyhedra is available, i.e., the variables are R and (z/n) and the constant parameters obtained from the fit are R_0 and B . The uniform bond limit assumes that a series of structures can be described by a single (R_0, B) pair. This is not the case in the authors’ analysis, since R_0 and B are themselves treated as the variables defining the correlation.

I do not question the observed linear correlation between B and R_0 . However, I do not believe that this linear relationship is correctly derived from the uniform-bond limit.”

Response. The referee’s last sentence is the crux, so we take it first. The relationship is derived exactly. For a shell with every bond at $\bar{R}$, Eq. (1) and the sum rule give $n \exp[(R_0 - \bar{R})/B] = z$, that is $R_0 = \bar{R} + B \ln(z/n)$, for every (R_0, B) that reproduces the structure. The admissible pairs for one regular structure therefore form a line through $(\bar{R}, 0)$ with slope $1/\ln(z/n)$. The referee’s own observation, that infinitely many pairs satisfy the sum rule, is this line, and Eq. (5) is its slope. Nothing is assumed about a series of structures sharing one pair.

The prediction is where the per-structure fits must fall. Each structure is fitted alone and, if nearly regular, is confined to its own line. The lines of different structures at fixed species and n are parallel and offset by the spread in $\bar{R}$, about $0.1 \text{ \AA}$, while the fitted points spread along them over ångströms or more (Response Figure 2), so a regression across structures recovers the common slope. Manuscript Fig. 1 tests that slope for 150 species and coordination pairs. The second-order terms of Eq. (3) set where a distorted structure falls on its line and produce the scatter treated in the near-pole analysis.

The referee’s reading is the conventional one, with $\bar{R}$ and z/n as the variables and one pair fitted across structures. The two uses do not conflict. They are the same equation read along different directions. Conventional fitting picks one underdetermined point on the line, which is why B has usually been fixed. We treat the direction of the line as the physical content, because Eq. (5) predicts it with no adjustable constant.

Changes. The Parameter-free slope section now states, after Eq. (5), “Equation (5) does not prescribe a single (R_0, B) for a species. It fixes the direction along which the joint fit of one structure is degenerate, so the per-structure pairs of a species at fixed n trace a line of predicted slope and the second-order terms set where each structure falls on it.”

Block 7 — (c) the B – R_0 correlation, published variations of B , and structural validity.

“A final remark concerns the correlation between R_0 and B . In the authors’ previous publication (Ref. 19 in the manuscript), it is shown that the ranges of variation of R_0 and B are still quite large. For instance, for Fe–O, R_0 varies from 0 to 4 Å, while B varies from approximately -2 to $+2$ Å. In these figures the data is shown as well as the fit. I find these variations very surprising. Deviations from the commonly used value of $B = 0.37$ Å have been investigated, for example, for Pb–O bonds by Krivovichev and Brown (Z. Kristallogr. 216 (2001), 245–247), who found that the Pb–O bonds were adequately described using $B = 0.49$ Å. This variation is much smaller than those reported by the authors.

I therefore wonder whether the structures used in the fits have been carefully checked for their physical and structural validity. If the valence-sum rule is imposed on structures that are themselves physically unrealistic, I imagine that the resulting bond-valence parameters could be expected to show large deviations from the usual values. This possibility may be particularly relevant when working with databases containing structures obtained from simulations rather than from experimentally well determined crystal structures.”

Response. The variations the referee quotes and the variations in the published literature are different quantities. The published values, including the $B = 0.49$ Å of Krivovichev and Brown for Pb–O, are pooled per-pair fits, in which one pair (R_0, B) is chosen for all structures containing the pair. Such a fit selects one point in a shallow valley of the residual surface whose floor runs along the line of Eq. (5), and the choice between 0.37 and 0.49 Å is a choice within that valley. The per-structure values in Ref. [19] and in the present work are the individual points along the valley floor. Their spread along the line is the degeneracy discussed under Blocks 5 and 6 and is expected to be large. Their spread perpendicular to the line is the quantity that measures the quality of the fit, and it is what the RANSAC residual threshold, the bootstrap slope errors (median relative error 0.2%) and the $R^2 \geq 0.3$ acceptance control. The species-level quantity comparable to a published B is the characteristic B^* , and the Supplemental Material compares the B^* distribution with the Brown–Altermatt, softBV and Gagné–Hawthorne sets. The mean offset from the Gagné–Hawthorne values on matched species is -0.01 Å.

On structural validity, the corpus consists of DFT-relaxed structures retrieved from the Materials Project with an energy-above-hull window of 0 to 50 meV/atom, so every structure lies within 50 meV/atom of the convex hull. Hull energies are recorded for 31,446 of the fitted shells, all within that window, with a median of 13 meV/atom. Unrealistic shells move points off the predicted line, whereas the large values occur on the line, and the near-degenerate shells that produce them are the most regular polyhedra in the corpus, not the least.

Changes. The Data provenance section of the Supplemental Material now states the stability filter: structures were retrieved from the Materials Project with an energy-above-hull window of 0 to 50 meV/atom, so every structure lies within 50 meV/atom of the convex hull.

Block 8 — partition thermodynamics and the expansion point.

“The authors introduce a thermodynamics-based formulation based on the similarity between the empirical bond strength expression and the classical partition function in thermodynamics. Despite this mathematical analogy, I believe that a physical argument is missing. I agree that this approach can capture the asymmetry of an atomic environment, but I do not see why the average distance obtained from this ‘thermodynamic average’ should be chosen as the center point of the Taylor expansion.

In fact, in my understanding, if the purpose of the expansion is to recover the empirical bond-strength expression, the relevant expansion point should be the value R_0 that best describes the valence-sum rule. It is at this point that the approach based on the Yukawa potential is identified with the empirical bond-valence expression.”

Response. The argument is physical, and we have now stated it in the Partition thermodynamics section where $R' = m_i$ is adopted (clean manuscript p. 5, marked-up diff p. 5). It has two parts.

First, the expansion point must lie within the interval $[R_{\min}, R_{\max}]$ spanned by the shell’s bonds, so that the error $\delta_{ij} = O[(R_{ij} - R')^2/(\lambda + R')^2]$ is set by the shell’s distortion and not by its distance from an external reference. R_0 generally does not lie there. By Eq. (10) it is offset from the centroid m_i by $B(\ln q_i - H_i)$, which is $B \ln(z/n)$ in the uniform limit, so it sits below the shell for $z < n$ and above it for $z > n$. The corpus confirms this. For $z < n$, R_0 lies below $R_{\min}$ in 70% of 42,071 shells, with a median offset of 0.27 Å against a median bond-length range of 0.18 Å, and for $z > n$ it lies above $R_{\max}$ in 64%. Expanding about R_0 would make the error depend on z/n rather than on distortion.

Second, within that interval the Gibbs mean is singled out by the weights. The linear term of Eq. (3) is kept exactly, so the shell normalization is second-order accurate for any R' , and the leading error is $-\sum_j p_{ij}(R_{ij} - R')^2/[2(\lambda + R')^2]$. The weighted second moment equals the shell variance plus $(m_i - R')^2$, minimized at $R' = m_i$ and nowhere else, where the error reduces to the distortion alone. We verified this numerically with exact Yukawa weights on distorted, asymmetric and near-degenerate model shells. The error minimum lies at m_i and the closed form holds to within 5%.

The identification the referee describes occurs at a later step. Eq. (10) identifies R_0 as the entropy-corrected centroid, and at the species level the inversion for λ^* is evaluated at $R' = R_0^*$, which is what the referee proposes.

Changes. The Partition thermodynamics section now closes, after $R' = m_i$ is adopted, with: “The choice is physical rather than formal. The expansion controls its error through the bond-length spread of the shell, so R' must lie within the interval spanned by the bonds, and R_0 generally does not. By Eq. (10) it differs from m_i by $B(\ln q_i - H_i)$, which in the uniform-bond limit is $B \ln(z/n)$, placing R_0 below the shell for $z < n$ and above it for $z > n$. Among lengths within the shell, m_i minimizes the valence-weighted second moment $\sum_j p_{ij}(R_{ij} - R')^2$, which equals the weighted variance of the shell plus $(m_i - R')^2$, so the leading linearization error, $-\sum_j p_{ij}(R_{ij} - R')^2/[2(\lambda + R')^2]$, is set by the distortion of the shell alone.”

Block 9 — closing remarks on the spread of (R_0, B) and transferability.

“After carefully reading the manuscript, I remain unconvinced by several aspects of the work. In particular, the R_0 and B values obtained from the fits, as shown in Figures 2S,

span ranges that appear to be physically difficult to justify. As I stated before, this probably can be an error in the generation of the figures, but in a previous publication of the same authors the variation shown by B was still large to me. More generally, if the variations of B and R_0 for a given interaction pair are indeed as large as those reported by the authors, this raises a more fundamental question concerning the usefulness of the bond-valence formalism. One of the strengths of bond-valence theory is that, for a given interaction $A-X$, a single pair of parameters (R_0, B) can be used to describe bonds in different structural environments. This transferability is what makes it possible to calculate bond valences from the observed bond distances and then evaluate the deviation from the valence-sum rule. The resulting bond-valence mismatch can, for example, provide useful information about local structural strain or possible ion-diffusion pathways.

If, instead, each individual structure or local environment requires its own (R_0, B) pair, it is not clear to me how the bond-valence mismatch should be defined.

*I may have misunderstood some aspects of the bond-valence formalism, as I am not an expert in this field. However, based on my reading of the manuscript, I cannot recommend the publication of the manuscript in *Physical Review Materials*.”*

Response. Transferability relates to the point made at the outset. Conventional parameters are transferable among the structures that satisfy the theory’s assumptions, and curated reference sets select those structures in advance.

We believe that two conflated meanings of (R_0, B) are the major source of the referee’s concerns. Conventional parameter sets assign one pair per cation–anion pair by pooling every structure that contains it into a single fit. Our per-structure (R_0, B) are a different object, one point per structure on a line whose slope the theory predicts, and they are not meant to be transferable. The quantity in our work that is analogous to a conventional parameter pair, and the most transferable one, is the characteristic pair (R_0^*, B^*) of Eq. (7), a single species-level constant obtained by aggregating the per-structure fits across the corpus. Bond valences and the bond-valence mismatch are computed from observed bond lengths with (R_0^*, B^*) exactly as with any published pair, so the diagnostic use the referee describes is unchanged. It is directly comparable to the published sets, and the manuscript says so. “The screening-branch B^* values are not universal but cluster near the conventional $B = 0.37 \text{ \AA}$ [8], and their comparison with published softness parameterizations [2, 10, 9] is given in the Supplemental Material.” That comparison (Supplemental Material, “Comparison with published softness parameters”) places the B^* distribution against the Brown–Altermatt, softBV and Gagné–Hawthorne values and reports the matched-species offsets. We opted not to develop the comparison of bond-valence fitting methods further in the main text of this physics paper, in order to keep the emphasis on the physical underpinnings of the exponential form. The methodological comparison is the subject of separate work.

Closing remarks

We thank the Second Referee for a demanding report. The revisions it prompted make explicit what the manuscript left implicit, that the predicted slope describes a family of per-structure fits and that the extreme parameter values are the degenerate limit of that family rather than failures of the structures or of the fit, and they supply the physical argument for the expansion point that the referee found missing. We hope the revised manuscript and this response resolve the concerns.

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